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Review

Fundamental Biological Features of Spaceflight: Advancing the Field to Enable Deep-Space Exploration

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SUMMARY

Research on astronaut health and model organisms have revealed six features of spaceflight biology that guide our current understanding of fundamental molecular changes that occur during space travel. The features include oxidative stress, DNA damage, mitochondrial dysregulation, epigenetic changes (including gene regulation), telomere length alterations, and microbiome shifts. Here we review the known hazards of human spaceflight, how spaceflight affects living systems through these six fundamental features, and the associated health risks of space exploration. We also discuss the essential issues related to the health and safety of astronauts involved in future missions, especially planned long-duration and Martian missions.

INTRODUCTION

Since the first dog (Laika) was launched into space in 1957, studies of spaceflight biology and aerospace medicine have led to new discoveries of the unique biological responses to the space environment. A principal goal of the National Aeronautics and Space Administration (NASA) and other space agencies such as the Japan Aerospace Exploration Agency (JAXA), the European Space Agency (ESA), Roscosmos State Corporation for Space Activities (ROSCOSMOS), and the China National

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Space Administration (CNSA) has historically been to expand scientific knowledge for humankind's benefit (National Aeronautics and Space Act, 1958; Science Mission Directorate, 2020). This includes a greater understanding of the universe as well as repurposing space-relevant research to advance science and technology on Earth. For example, technology developed from space missions has led to the creation of camera phones, velcro, and artificial limbs (NASA Technology Transfer Program, 2019). Analogously, spaceflight biology creates a wealth of scientific knowledge and technologies which contribute to terrestrial, medical, and biological sciences.

A large proportion of spaceflight biology research addresses the risks of space environment on human health. Specifically, two major health hazards that are related to spaceflight are space radiation and microgravity. Space radiation is mainly composed of high-energy protons produced from solar particle events (SPEs), heavy ions contained in galactic cosmic rays (GCRs), and secondary particles resulting from interaction with spacecraft shielding (Chancellor et al., 2014). Notably, the International Space Station (ISS) circumnavigates the planet in low Earth orbit (LEO), as have all previously crewed missions except Apollo. The missions within LEO benefit from protective effects of Earth's magnetosphere (Hassler et al., 2014), but upcoming exploration missions to the Moon, Mars, and beyond will expose crewmembers to increased GCR and likely high-intensity SPE. Furthermore, the health risks from space radiation are exacerbated by microgravity, limited contact with Earth, and physiological and psychological effects from extended isolated confinement in a closed and hostile ecosystem (Figure 1).

In this review, we outline the fundamental biological features associated with spaceflight that are related to human health risks and aerospace medicine (Figure 2). These features can serve as a guide for future fundamental and applied spaceflight biology research, including personalized medical therapies for astronauts and countermeasure development. We also list open-access spaceflight biological data and biospecimen resources available for request by the scientific community.

HAZARDS OF HUMAN SPACEFLIGHT

The fundamental biological features associated with the space environment occur in response to multiple spaceflight hazards. The main spaceflight hazards that impact human biology (Figure 1) are described below.

Space Radiation

Exposure to ionizing radiation (Figure 1) is a major health risk for upcoming deep-space missions. The space-radiation environment includes energetic solar particles emitted during solar flares and coronal mass ejections, as well as GCRs, which are composed of electrons and positrons (2%), protons (85%), helium nuclei (12%), and heavier ions referred to as high-energy and high-charge particles (HZE; 1%) (Durante and Cucinotta, 2008). Although SPEs are sporadic and can be shielded by spacecraft (Nelson, 2016), the health impacts of chronic exposure to GCR, especially HZE particles, are of particular concern due to their ionizing nature, high penetrability, and high energy deposition (Niemantsverdriet et al., 2012). On the ISS, astronauts are partially protected by the magnetic field of the Earth but remain exposed to an average dose of 100–200 milliSievert (mSv) radiation/year (Cucinotta and Durante, 2006), whereas the annual exposure limit for professional radiation workers is 50 mSv/year (Cucinotta, 2010). Doses received during a future trip to Mars will be even higher, up to 350 mSv/year for an ~3-year

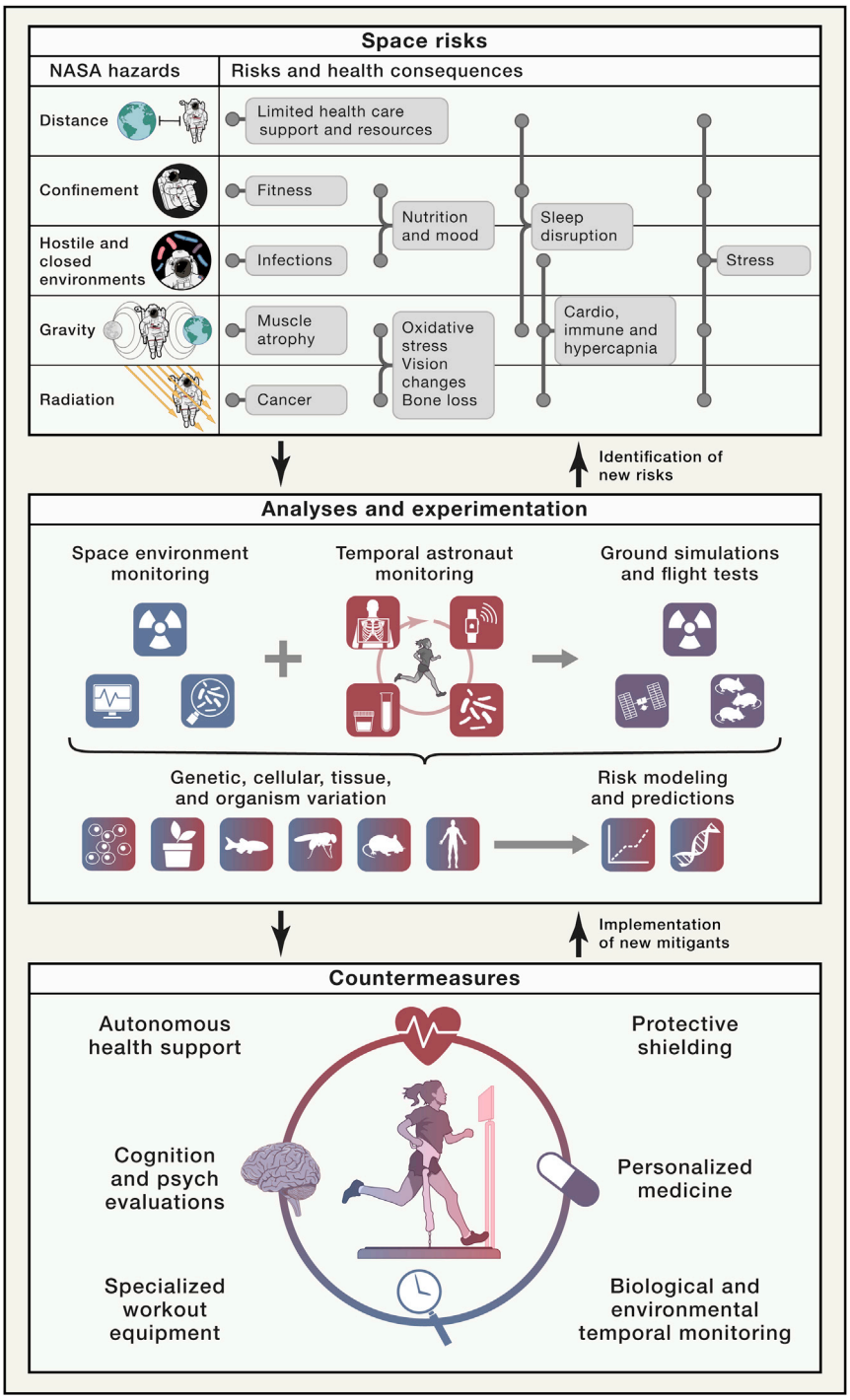


Figure 1. Information Flow from the Space Environment

The space environment introduces several different hazards (distance, confinement, hostile/closed environments, radiation, and microgravity), which have health risks and consequences that span multiple organ systems. Monitoring the space environment and astronauts help inform ground simulations and flight tests. These summative analyses and experimentation lead to the development of countermeasures precisely targeted toward the identified risks and health consequences posed by the hazards of the space environment. This information flow is a closed loop with feedback throughout multiple levels; for example, models could reveal new risks and the implementation of countermeasures might shift the models.

mission (Zeitlin et al., 2013; Iosim et al., 2019), and will primarily consist of highly biologically active GCRs. Therefore, understanding the biological effects of GCRs and their components is central for planning missions beyond the protection of Earth's magnetosphere. In addition, exposure to ionizing radiation during spaceflight occurs at very low dose rates (less than 1 mSv/day) (Zeitlin et al., 2013); therefore, more investigations are needed to understand the health effects caused by chronic, low-dose-rate radiation exposure (Acharya et al., 2019).

Microgravity and Altered Gravity

All life on Earth has evolved and adapted to a single downward vector of gravity. In the space environment, this vector is mostly removed, resulting in microgravity conditions (Figure 1). Also, there is a range of varying gravity levels on other planetary bodies that will be experienced by space travelers, such as on the Earth's Moon (1/6th gravity [G]) and Mars (1/3rd G). Additionally, there are often intense forces experienced during exit and descent transitions through atmospheres, at many times the force of gravity (3–6 G), which will subject biological systems to stress. Microgravity induces cellular and molecular adaptations and changes in the genome, epigenome, as well as the proteome, and these changes create risks for a range of pathologies (Demontis et al., 2017). At the organismal level, biological processes that have evolved to respond to gravity can cause abnormal physiological responses in microgravity. For example, fluids, including blood, shift upward toward the head and thorax, leading to decreased leg volume and compensatory cardiovascular system changes (Hughson et al., 2018). Bone and muscles also no longer experience gravitational loading, and they atrophy through remodeling processes, whose molecular mechanisms are only partially understood. In addition, synergistic effects of prolonged microgravity in combination with other space environment hazards may exacerbate complex health problems in astronauts embarking on long-duration missions (Jillings et al., 2020).

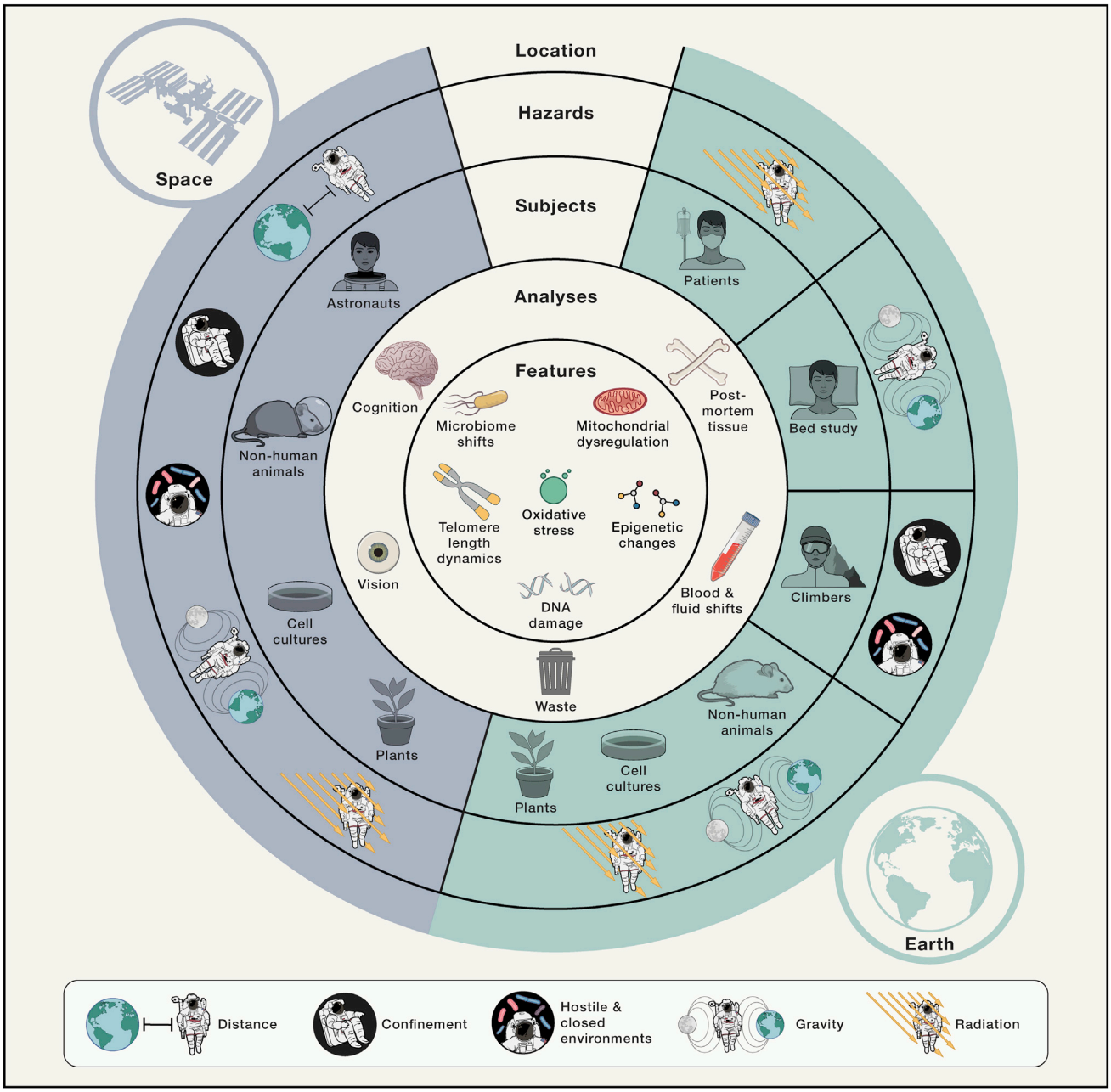


Figure 2. Biological Features of Spaceflight

Experiments and analyses have been designed both on Earth and in space to study the impacts that spaceflight has on human biology and physiology. Subjects vary from plants, cell cultures, non-human animals, and humans. In space, the hazards outlined in Figure 1 are highlighted here as they pose health risks and drive many physiological changes seen during spaceflight. On Earth, these hazards are simulated in different experiments and milieus. For example, bed studies, radiation of cancer patients, and climbers help to mimic some of the hazards experienced in the extreme environment of space and provide some insights on their impacts on human physiology. These experiments involve several analyses across multiple systems including cognition, vision, waste, blood, fluid, and post-mortem tissue measurements. Collectively, these experiments and analyses reveal significant recurrent molecular and cellular features of spaceflight including DNA damage, oxidative stress, mitochondrial dysregulation, microbiome shifts, epigenetic changes, and telomere length dynamics. These features help to understand what drives some of the systemic and physiological impacts of spaceflight.

Confinement and Isolation

Deep-space travel will confine and isolate astronauts in a limited-size spacecraft for prolonged periods of time. Extended physical and social isolation, along with unprecedented distance from Earth, will likely increase psychological, behavioral, and physiological health risks (Pagel and Choukèr, 2016). Social isolation (Figure 1) has been studied on Earth by using analog environments on human subjects (Basner et al., 2014) as well as animal

models, demonstrating that it might lead to neurological deficits, including impaired hippocampal functions (Leser and Wagner, 2015) and increased anxiety (Wallace et al., 2009). In addition, based on human results from analog missions in the Arctic, isolation may directly contribute to immune dysregulation (Crucian et al., 2014a). Social isolation combined with other spaceflight hazards could worsen outcomes, as suggested by findings in animal models exposed to simulated microgravity combined with social isolation (Tahimic et al., 2019).

Hostile and Closed Environment

Continuous and prolonged habitation in an enclosed spacecraft ecosystem presents a biologically hostile and closed environment for astronaut health. Spacecraft habitability must be monitored for temperature, air quality, microbial inhabitants, pressure, lighting, and noise to help ensure effective countermeasures for a healthy environment. Constant noise, high carbon-dioxide levels, and limited microorganism ecosystems together might affect cardiovascular, neurological, and immune health. For example, levels of ambient noise have been demonstrated to contribute to cardiovascular impairments, sleep disturbance, and cognitive deficits (Münzel et al., 2020). In addition, due to limited efficacy of air recycling systems, increased CO_2 is a common feature on spacecraft and can lead to a hypoxic/hypercapnic response (Beheshti et al., 2018). Finally, prolonged confinement is likely to reduce the variability of the environmental microbiome, which might adversely affect astronaut immune functions and metabolism (Voorhies and Lorenzi, 2016).

Distance from Earth

The final major spaceflight hazard is distance from Earth itself (Longnecker and Molins, 2006) (Figure 1), which causes psychological stress and disrupts team dynamics (Stuster, 1996). Distance from Earth additionally will constrain medical treatment options and other equipment capabilities due to communication delays, limited medical capabilities on board, and no rapid evacuation nor immediate rescue during missions beyond LEO. Thus, there is an unmet need to develop autonomous health support for the flight medical officers, which could include surgical interventions (Panesar and Ashkan, 2018), further development in health monitoring and wearable sensors (Bellisle et al., 2020), artificial-intelligence-assisted medical diagnostics, on-board genetics and sequencing capacity (Castro-Wallace et al., 2017; McIntyre et al., 2019), and health-risk prediction models (Chancellor et al., 2018; Scott et al., 2020).

CHALLENGES OF SIMULATING AND MODELING THE SPACE ENVIRONMENT

Since the beginning of human spaceflight, NASA and international partners have been monitoring astronaut health and sending biological payload experiments to space. Besides human research, various organisms have flown in payloads to characterize spaceflight biological effects: sea urchins, quail, stem cells, soybean, bacteria, jellyfish, *Arabidopsis thaliana*, nematodes, yeast, fruit flies, rats, newts, radishes, and mice. Although recent space-launch payload delivery systems have reduced costs, the expenditure of sending humans to space and delivering spaceflight payload experiments is still somewhat extraordinary compared to Earth-based experiments. Additionally, in order to advance the field of knowledge, the scientific evidence is constrained by limited experimental sample sizes and an overall limited volume of spaceflight data and experiments. Thus, there is a need to augment these studies carried out in space.

Terrestrial spaceflight simulations, Earth environmental analogs, and ground investigations of spaceflight-mediated health impairments can accelerate the discovery of effective countermeasures and basic biological adaptation (Figure 1). Some hazards and stressors, such as social isolation and ambient noise, are comparatively easy to replicate in human analog environments (Nasrini et al., 2020), but microgravity and ionizing radiation are considerably more difficult to model on Earth. Although true microgravity is impossible to achieve, the vector of gravity can be dispersed by laboratory approximations, including rotating wall vessels, clinostats, and random positioning machines for cellular, plant, and tissue models (Grimm et al., 2018; Kiss et al., 2019). There are also suspension devices for animal models (Globus and Morey-Holton, 2016; Mortreux et al., 2018) and simulating effects on blood flow in space by using bed rest for human subjects (Koppelmans et al., 2017). Simulated microgravity in humans and rodents has been successfully combined with other space-relevant stressors, such as increased CO_2 (Lee et al., 2019), ionizing radiation (Kwok et al., 2019), and social isolation (Tahimic et al., 2019).

The space-radiation environment is challenging to model on Earth because it involves chronic, prolonged exposure to mixed fields of ionizing radiation at low dose rates. It is typically approximated on ground by the same total dose of acute irradiation, such as gamma rays and X-rays. A recently developed method to simulate GCR in the NASA Space Radiation Laboratory (NSRL) (Simonsen et al., 2020) has shown promising results in cellular, tissue, and animal models (Malkani et al., 2020; Paul et al., 2020; Wuu et al., 2020). Currently available simulators of low-dose-rate irradiation include gamma ray flood sources (Bellone et al., 2016), a chronic neutron irradiation animal facility (Borak et al., 2019), and a series of low-dose, fractionated irradiations at NSRL.

SIX FUNDAMENTAL MOLECULAR AND CELLULAR FEATURES OF SPACEFLIGHT

Throughout the literature, myriad biological response features of spaceflight have been documented, and they arise from six basic changes at the molecular and cellular level. Here, we mapped the biological responses along their known biological mechanisms and related these changes to the phenotypic and physiological impact (Figure 2).

Oxidative Stress

The first feature of spaceflight biology is elevated oxidative stress (Figure 2), and it is one of the main responses to spaceflight conditions that can trigger DNA damage. Oxidative stress arises when the amount of free radicals exceeds the natural antioxidant capacities of the cell (Spitz et al., 2004). Exposure to space radiation, hypoxia, and microgravity enhances the cellular production of reactive oxygen and nitrogen species (ROS, RNS),

as measured in 13 astronauts in long-duration (>4 months) ISS missions (Goodwin and Christofidou-Solomidou, 2018) and 59 astronauts' 8-oxo-guanosine levels in urine (da Silveira et al., 2020). Focusing on radiation-induced oxidative damage, an linear energy transfer (LET)-dependent oxidative stress response has also been observed in human immune cells *ex vivo* (Pariset et al., 2020a). At the physiological level, oxidative stress and oxidation-reduction imbalance contributes to spaceflight-related dysregulations of the cardiovascular, immune, neurological, and metabolism systems.

DNA Damage

The second feature is DNA damage (Figure 2). DNA damage is created by ionizing radiation that induces DNA breaks, either through direct interactions with the DNA molecule or through indirect radiolysis (Hall and Giaccia, 2006). Although single-strand breaks are easily repaired through excision repair mechanisms (Hall and Giaccia, 2006), double-strand breaks (DSBs) involve more complex repair processes. Risks of misrepair can elicit cell-cycle arrest, cell death, mutations, chromosomal rearrangements (Costes et al., 2010; Sridharan et al., 2015), and subsequent carcinogenesis (Tang et al., 2015).

Unlike low-LET radiation, HZE particles in the space environment deposit their energy along densely ionizing tracks, traversing thousands of cells and delivering elevated local doses in the cell nucleus (Magee and Chatterjee, 1980). This distinct energy-deposition pattern is responsible for high relative biological effectiveness (RBE) for carcinogenesis, especially for high-LET GCR components such as ^{12}C, ^{20}Ne, ^{28}Si, ^{48}Ti, and ^{56}Fe (Cacao et al., 2016). Indeed, DSBs generated by HZE particles cluster into radiation-induced foci (RIF) (Chiolo et al., 2011) and tend to repair more slowly than do individual DSBs induced by low-LET radiation (Pariset et al., 2020b). Because a single RIF can contain multiple breaks, its interpretation is complex and takes into account the capacity of damage recognition, the repair efficiency, the complexity of damage, and the function of the labeled protein (Costes et al., 2006; Han et al., 2006).

DNA damage was recently demonstrated for the first time in astronauts, based on directional genomic hybridization that detected an increase of intra-chromosomal inversions during and after spaceflight, a finding potentially related to stem-cell damage, clonal hematopoiesis, and/or instability (Garrett-Bakelman et al., 2019; Luxton et al., 2020a, 2020b; Trinchant et al., 2020). Studying extremely radio-tolerant organisms, such as the tardigrade, has revealed a unique damage suppressor protein that potentially could help protect against DNA damage resulting from chronic exposure to space radiation (Westover et al., 2020 [unpublished data]).

Mitochondrial Dysregulation

The third major feature of biological responses to spaceflight is mitochondrial dysfunction, which is also heavily linked to oxidative stress (Figure 2). Mitochondrial dysfunction can be indicated by a reduction in the expression of nuclear DNA (nDNA)-coded mitochondrial oxidative phosphorylation (OXPHOS) genes with a marked compensatory induction of mtDNA-coded OXPHOS genes. Mitochondrial OXPHOS and associated fatty-acid oxidation has recently been shown to be impaired in astronauts, which is consistent with their elevated systemic lipid levels (Garrett-Bakelman et al., 2019). Additional data from mice flown to space have also shown that the liver function directly regulates carbohydrate and lipid levels (da Silveira et al., 2020). Thus, changes in systemic mitochondrial function that alter substrate oxidation should alter substrate levels and liver function. Further evidence of mitochondrial impairment in astronauts include increased mitochondrial ROS production associated with elevated 8-hydroxydeoxyguanosine (a product of DNA oxidation) and prostaglandin levels, reduced antioxidant defense mechanisms, and altered circadian rhythms with reduced NAD+/NADH ratios impairing Sirt1 deacetylation of BMAL (Nakahata et al., 2008). In fact, a multi-omic analysis of human and mouse responses to spaceflight has revealed that one of the most important physiological alterations is mitochondrial dysregulation (da Silveira et al., 2020).

Ionizing radiation induces ROS not only in cellular cytoplasm, but also in mitochondria, which occupy a significant cellular area and are thus a common target of radiation impact (Leach et al., 2001). Amplification of mitochondrial ROS by ionizing radiation leads to mutations in mitochondrial DNA and subsequent disturbance of the expression of critical proteins for mitochondrial and cellular functions (Azzam et al., 2012). Notably, oxidative damage in mitochondrial DNA (mtDNA) is several-fold higher than in nuclear DNA (Richter, 1992), which can be due to the lack of protective histone proteins and less-efficient DNA repair mechanisms for mtDNA (Wiseman and Halliwell, 1996). In addition, a significant decrease in the ratio between mtDNA to nuclear DNA was observed between pre- and post-flight astronaut hair samples, as well as a significant decrease in the expression of genes related to oxidative stress (Indo et al., 2016), and more in-flight mtDNA and mtRNA were observed in peripheral blood during NASA's first year-long mission (Garrett-Bakelman et al., 2019). Microgravity could also yield another source of mitochondrial oxidative stress independent from the effects of radiation (Zhang et al., 2014b).

Of note, astronauts are exposed to chronic low-level radiation from a wide spectrum of ionizing particles, including GCR. In addition to breaking the mtDNA, ionizing radiation creates mitochondrial ROS, which damages mtDNA, proteins, and lipids. The iron sulfur centers within mitochondrial OXPHOS complexes I, II, and III and mitochondrial aconitase are acutely sensitive to ROS, resulting in impaired electron transport and further increased ROS (Melov et al., 1999). Mitochondrial ROS also modulates the function of the DNA damage protein ATM, linking DNA damage and mitochondrial function (Sharma et al., 2014). Given that impaired mitochondrial function is a consistent consequence of spaceflight, it is possible that these alterations account for some aspects of health risks.

Epigenetic and Gene Regulation Changes

The fourth feature of spaceflight biology is epigenetic (including gene regulation) changes. Of note, many epigenetic and gene-expression changes were found in Scott Kelly's year-long mission (Garrett-Bakelman et al., 2019), but the majority of them reversed to normal upon return to Earth, and some sites were even less altered in comparison with the ground control (his twin brother Mark Kelly, during the same amount of time). For example, the overall DNA methylation landscape of Scott was less diverged than Mark's, even after a year in space. This

indicates that a tight, reliable regulation of sleep, food intake, daily exercise, and engagement with the environment on the ISS could actually be a constraint on epigenetic changes. Nonetheless, other works on chromatin landscapes from sorted cells and single-cell sequencing (scATAC-seq, scRNA-seq) have shown significant differences in chromatin accessibility and epitope levels before and after spaceflight (Gertz et al., 2020; Malkani et al., 2020).

In particular, chromatin decondensation after DSB induction (Krawczyk et al., 2012) improves accessibility of DNA repair proteins (Chiolo et al., 2011) and homology search during repair (Miné-Hattab and Rothstein, 2012). As a consequence, chromatin decondensation is a major mechanism of radiation-induced epigenetics by promoting gene transcription (Handy et al., 2011). In addition to DSBs, ROS were shown to directly modulate chromatin condensation (Coluzzi et al., 2019), which indicates that both radiation- and microgravity-induced (Adrian et al., 2013) oxidative stress could induce epigenetic changes through chromatin relaxation.

Telomere-Length Dynamics

The fifth feature of spaceflight is represented by changes in telomeres. The ends of human chromosomes are capped by telomeres, tandem arrays of repetitive G-rich sequences and associated proteins that serve to protect chromosomal termini from degradation. Telomeres also preserve genome stability by preventing natural chromosomal ends from being recognized as DSBs and triggering inappropriate DNA damage repairs (DDRs). Due to the "end-replication problem," telomere length erodes with cell division and thus with aging. Oxidative stress, infection, and inflammation also contribute to telomere shortening, as do a host of lifestyle factors including stress (e.g., nutritional, physical, psychological) and environmental exposures (e.g., air pollution, radiation). Therefore, telomere-length dynamics (changes over time) represent a relevant and integrative biomarker for astronauts, because they could reflect the combined exposures and experiences encountered in the extreme environment of space. Furthermore, longitudinal analysis of telomere length provides an informative indicator of general health, aging, and aging trajectories, because altered telomere length is linked to age-related pathologies, including dementia, cardiovascular disease, and cancer, all conditions with the potential of influencing astronaut health and performance during and after long-duration missions.

The question of aging associated with long-duration spaceflight was addressed for the first time in astronauts by assessing telomere-length dynamics and genome stability in space- and Earth-bound twins Scott and Mark Kelly (Garrett-Bakelman et al., 2019). In similar studies, blood samples were collected before, during, and after spaceflight from a cohort of ten unrelated astronauts and age- and sex-matched ground-control subjects (Luxton et al., 2020a). The most striking observation was telomere elongation during spaceflight for all crewmembers for whom in-flight samples were available, regardless of mission duration, sample type, or means of measurement (Luxton et al., 2020b). Telomere length also shortened rapidly upon return to Earth, and, in general, crewmembers had many more short telomeres after spaceflight than they did before. Chronic exposure to the space-radiation environment and intimately associated DDRs, as well as a variety of stress responses, potentially contribute to telomerase-dependent and/or independent pathways of telomere maintenance (Bezdan et al., 2020; da Silveira et al., 2020; Grigorev et al., 2020 [unpublished data]; Luxton et al., 2020a, 2020b; Trinchant et al., 2020). Although definitive mechanisms and temporal health effects of such dramatic, spaceflight-associated shifts in telomere-length dynamics are yet to be determined, results highlight the importance of monitoring individual crewmember telomere length and genome stability as a means of evaluating general health, disease, and aging risk, information that then should be considered for development of personalized aerospace medicine and countermeasures for future astronauts.

Microbiome Shifts

The last feature revolves around spaceflight-mediated shifts in host-microbe interactions. The microbiome is a dynamic ecosystem of microorganisms that live in us, on us, and all around us, and its shifts are driven by several different factors. There are two main components of the microbiome to study in spaceflight biology. First, the microbiome of the astronauts themselves can change, spanning the gut, skin, oral, and other niche bodily microbiomes. Results from the NASA Twins study demonstrated a reversible shift in the *Firmicutes* to *Bacteroidetes* (F/B) ratio, but overall diversity in the gut microbiome was maintained (Garrett-Bakelman et al., 2019). Astronauts on 6- to 12-month missions also showed stable or significant increases in gut microbiome richness and diversity (Voorhies et al., 2019). Similarly, examination of feces from mice flown on the Space Shuttle and ISS revealed consistent spaceflight-associated shifts in the microbiome and increased F/B ratio with increased microbial diversity (Jiang et al., 2019). Despite these initial studies, there is still much not fully understood regarding spaceflight impacts on microbiome dynamics and health consequences (Voorhies et al., 2019), including salivary microbiome shifts and viral (e.g., herpes) reactivation (Urbaniak et al., 2020).

The second component is the microbiome of the spacecraft environment. Shotgun metagenome sequencing of the ISS environment has revealed the presence of viable microbial communities (Singh et al., 2018), revealing occasional increases in antimicrobial resistance and virulence gene factors. Comparative analysis of the microbial compositions of ISS with Earth analogs revealed that ISS environmental surfaces were different in microbial composition (Lang et al., 2017a; Singh et al., 2018). In addition, studies have shown that the ISS environmental microbiome most resembles the crew's skin microbiome (Avila-Herrera et al., 2020). Moreover, the whole-genome sequences of several hundred strains of the ISS environmental surface samples have led to the discovery of novel species. These include *Solibacillus kalamii* (Checinska Sielaff et al., 2017), *Kalamiella piersonii* (Singh et al., 2019), and *Bacillus creus-anthracis* clade (Venkateswaran et al., 2017).

SYSTEMIC AND PHYSIOLOGICAL HEALTH RISKS OF SPACEFLIGHT

The cellular and molecular responses of spaceflight highlighted above have significant physiological and systemic

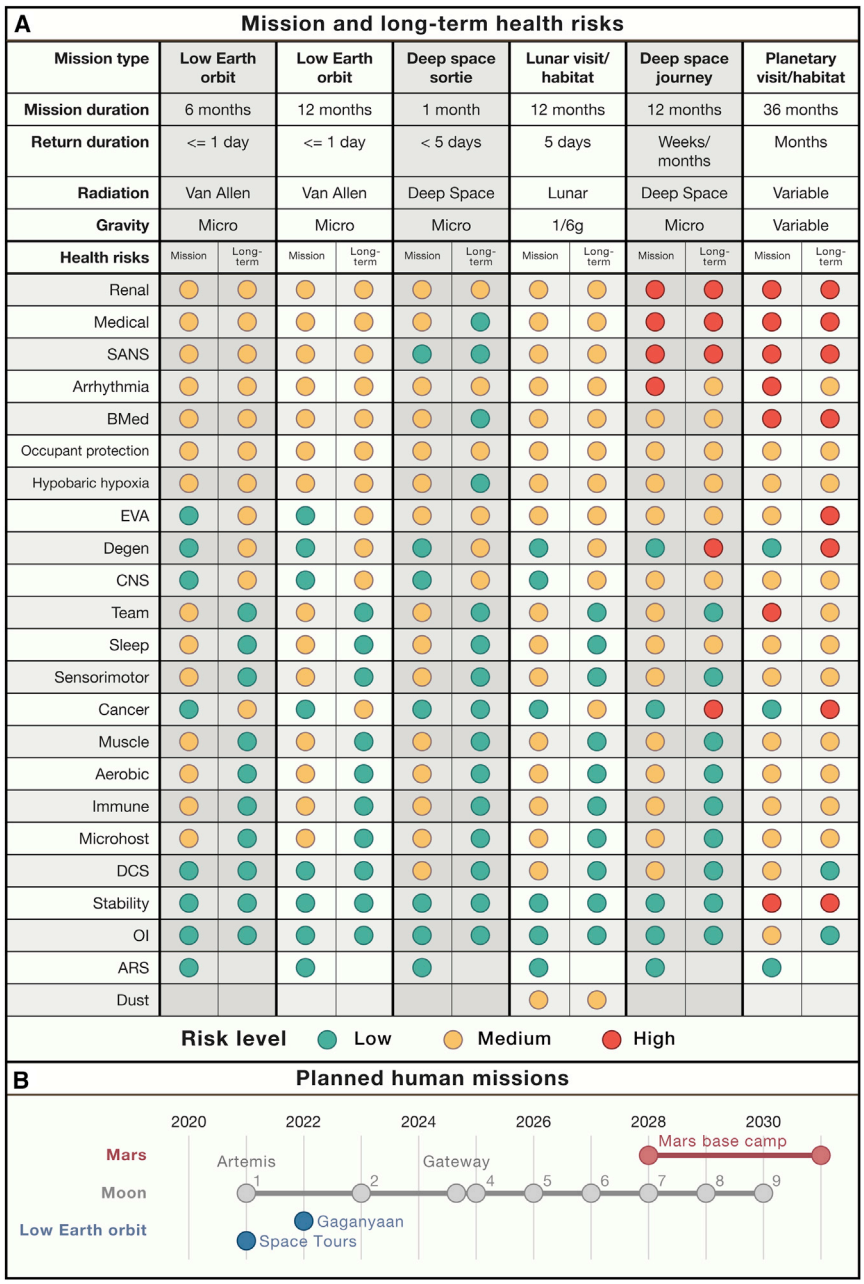


Figure 3. Astronaut Mission and Long-Term Health Risks

(A) Mission-specific and long-term health risks are outlined based on mission type. The mission types include LEO, deep-space sortie, lunar visits, deep-space journey, and planetary visits. These missions are defined by their characteristics including mission duration, return duration, radiation exposure, and gravity. Astronaut health risks spanning multiple organ systems are included with color scale to denote the NASA-defined overall safety risk scores (low to high, green to red, respectively) for mission risk and long-term risk. The NASA Human Research Roadmap (https://humanresearchroadmap.nasa.gov/Risks/) provides the guide for the health risks of interest. The risks are further defined as follows:

Aerobic, risk of reduced physical performance capabilities due to reduced aerobic capacity; arrhythmia, risk of cardiac rhythm problems; ARS, risk of acute radiation syndromes due to SPEs; BMed, risk of adverse cognitive or behavioral conditions and psychiatric disorders; cancer, risk of radiation carcinogenesis; CNS, risk of acute (in-flight) and late central nervous system effects from radiation exposure; DCS, risk of decompression sickness; degen, risk of cardiovascular disease and other degenerative tissue effects from radiation exposure and secondary spaceflight stressors; dust, risk of adverse health and performance effects of celestial dust exposure; EVA, risk of injury and compromised performance due to EVA operations; hypobaric hypoxia, risk of reduced crew health and performance due to hypobaric hypoxia; immune, risk of adverse health event due to altered immune response; medical, risk of adverse health outcomes and decrements in performance due to inflight medical conditions; microhost, risk of adverse health effects due to host-microorganism interactions; muscle, risk of impaired performance due to reduced muscle mass, strength, and endurance; occupant protection, risk of injury from dynamic loads; OI, risk of orthostatic intolerance during re-exposure to gravity; renal, risk of renal stone formation; SANS, risk of Spaceflight-Associated Neuro-ocular Syndrome; sensorimotor, risk of impaired control of spacecraft/associated systems and decreased mobility due to vestibular/sensorimotor alterations associated with spaceflight; sleep, risk of performance decrements and adverse health outcomes resulting from sleep loss, circadian desynchronization, and work overload; stability, risk of ineffective or toxic medications due to long-term storage; team, risk of performance and behavioral health decrements due to inadequate cooperation, coordination, communication, and psychosocial adaptation within a team.

(B) Future planned human missions to LEO, the Moon, and Mars over the next decade until 2030. ISS missions are planned throughout all years in LEO.

implications regarding astronaut health. As outlined in Figure 3, the health risks span several different organ systems and are characterized by risk to the astronauts during their mission, as well as by long-term health risks post-mission. Here we review the most researched and critical organ systems impacted by spaceflight and associated health risks. These include the cardiovascular system, central nervous system, increased cancer risks, the musculoskeletal system, immune system, gastrointestinal system, dysregulation of the circadian system, and vision changes.

Cardiovascular Dysregulation

Spaceflight removes normal loading effects from gravity on the cardiovascular system and initiates "aging-like" deconditioning, including loss of physical fitness, arterial stiffening, and development of insulin resistance. Acute effects on cardiovascular

physiology include decreased circulatory blood volume, reduction in arterial blood diastolic pressure, reduced left-ventricular size, post-flight orthostatic intolerance, and reduced cardiac contractility (Patel, 2020; Walls et al., 2020 [unpublished data]). Space radiation compounds the effects of microgravity as seen in the heart and vasculature, along with DNA damage and oxidative stress (Patel, 2020; Bishawi et al., 2020 [unpublished data]) (Figure 3A).

Orthostatic intolerance has commonly occurred post-flight, with many astronauts experiencing syncope and tachycardia (Buckey et al., 1996). Space radiation in mouse models causes aortic stiffness (Soucy et al., 2011) and accelerated atherosclerosis (Yu et al., 2011), along with increased DNA oxidation, myocardial fibrosis, and modified cardiac function (Yan et al., 2014). Small vessels also respond to space radiation: there is a loss of hippocampal microvasculature after exposure to iron ions (Mao et al., 2010), and, similarly, in human 3D micro-vessel models, heavy ions cause micro-vessels to collapse (Grabham et al., 2011; Malkani et al., 2020).

In addition, space radiation also inhibits angiogenesis in human micro-vessel models and does so by two distinct mechanisms depending on whether the radiation is composed of light or heavy ions (Grabham et al., 2013). Light ions primarily inhibit the early motile stages of micro-vessel formation, whereas heavy ions prevent the later stages of development, although they also act synergistically (Wuu et al., 2020). In small blood vessels, the inhibition of angiogenesis causes a gradual loss of vessels because of the lack of replacement of damaged vasculature (rarefaction), which leads to pathologies in the deprived tissues (Vernice et al., 2020). On an organismal level, rarefaction eventually causes pain in the extremities, high blood pressure, a long-term increase in the risk of hypertension, arterial thromboembolism, cardiac ischemia, and cardiac dysfunction (Abdel-Qadir et al., 2017). Because micro-vessels in the central nervous system (CNS) are also damaged by the space environment, they can also contribute to the CNS impairments described below.

CNS Impairments

CNS damage, caused by prolonged GCR exposure beyond Earth's magnetic field, is a recognized critical health risk for upcoming missions (Figure 3A). Current knowledge of GCR neurotoxicity is based on rodent ground-simulation models (Simonsen et al., 2020). These simulated GCRs and their most biologically active components, HZE particles, induce cognitive and behavioral deficits (Krukowski et al., 2018) correlated with neuroinflammation and neuronal damage (Parihar et al., 2018). These physiological and behavioral impairments continue over a prolonged period after GCR irradiation (Parihar et al., 2018), suggesting the possibility of additional neurological damage during post-flight re-acclimation to ground conditions.

Spaceflight, simulated microgravity, and GCR irradiation in rodent and other model organisms all share the same CNS effects: inflammation and oxidative stress at the cellular-tissue level (Santucci et al., 2012), along with behavioral deficits (Acharya et al., 2019; Bellone et al., 2016). Thus, during deep-space missions, combined spaceflight stressors could be expected to compound CNS damage. This was recently shown through executive function impairments, induced by protracted neutron exposure and exacerbated by sleep fragmentation (Britten et al., 2019).

Based on animal models, CNS responses to spaceflight and simulated GCR irradiation resemble various terrestrial disorders, including normal neurological aging and neurodegeneration (Clément et al., 2020). Therefore, GCR exposure might exacerbate aging-related brain damage. On the other hand, the mechanistic similarities between spaceflight and neurological diseases encourage the use of recently developed computational tools for drug repurposing (Nelson et al., 2019) to discover novel therapeutic targets and FDA-approved countermeasures.

CNS impairments are expected to overlap with systemic effects of spaceflight, including immune dysfunction (Mehta et al., 2017), which might exacerbate neuroinflammation via the blood-brain barrier. This interaction between CNS and systemic immunity is supported by the discovery that the peripheral immune phenotype can serve as a biomarker for behavioral deficits after HZE particle irradiation (Krukowski et al., 2018). Thus, it will be crucial to evaluate the blood-brain barrier both as a target of CNS damage (Bellone et al., 2016) and as a potential venue for delivering systemically applied countermeasures into the brain (Kariolis et al., 2020).

Increased Cancer Risk

Radiation-induced cancer is one of the main risks of deep-space exploration (Figure 3A), and space agencies around the world have set career dose limits based on the pojected risk of developing cancer. As an example, NASA has set 1Sv as the career limit for a 25-year old female, ranging up to 4Sv for a 55 year-old male (https://www.nasa.gov/sites/default/files/atoms/files/space_radiation_ebook.pdf), where a dose of 1 Sv is associated with a 5% increased risk of dying from cancer. Except for particle-based radiotherapy-induced secondary tumors (Chung et al., 2013), there is no equivalency on Earth to evaluate cancer risk from GCR components. For this reason, epidemiological studies of atomic bomb survivors exposed to acute doses of low-LET radiation have been scaled to anticipate cancer risk from high-LET radiation by using the RBE concept (Cucinotta and Durante, 2006), despite the different energy deposition pattern between photons and HZE particles (Costes et al., 2006).

Due to the similarity between rodent and human radiation-induced cancer incidence (Storer et al., 1988), simulations of GCR components on Earth have been extensively applied to rodent models, showing higher induction of some tumors compared to models of low-LET radiation (Cucinotta et al., 2013). Specifically, high RBE of HZE particles has been reported for induction of mammary tumors (Illa-Bochaca et al., 2014), Harderian gland tumors (Kennedy et al., 2008), skin tumors (Burns et al., 2007), intestinal colorectal cancer (Datta et al., 2013), leukemogenesis (Peng et al., 2009), and hepatocellular carcinoma (Weil et al., 2014). On the other hand, HZE ions are no more effective than are gamma rays for inducing acute myeloid leukemia (Weil et al., 2014) and ovarian cancer (Watanabe et al., 1998), which might indicate different underlying mechanisms for induction of these tumor types. Importantly, no novel tumor types have been observed in rodents exposed to HZE ions in comparison with spontaneous and low-LET-induced tumors (Bielefeldt-Ohmann et al., 2012). Additional data are

required to better understand the effect of dose fractionation on carcinogenesis, given current inconsistencies (Bielefeldt-Ohmann et al., 2012; Burns et al., 2007; Weil et al., 2014).

In humans, cancer risk predictions have been mostly based on computational models of the interactions between GCR particles and biological targets (Cucinotta et al., 2013). Recently, human responses to GCR components have been evaluated *in vitro* in a large cohort of 600+ healthy donors to take into account inter-individual variabilities (Pariset et al., 2020 [unpublished data]). The emergence of novel 3D tissue models (Low and Giulianotti, 2019) and comprehensive systems biology analysis (Barcellos-Hoff, 2008) will provide additional input on human responses to real spaceflight conditions to improve cancer risk predictions.

Muscle Degeneration

Exposure to microgravity induces loss of muscle mass, volume, and performance (Figure 3A). This effect occurs especially in the legs, on both short- and long-duration flights (Fitts et al., 2000). As with bone, regional changes in muscle loss appear to be dependent on the role of each region in counteracting gravity, and although the lower extremities are significantly affected, the muscles of the neck are not (McNamara et al., 2019). Biochemical profiling of astronauts before, during, and after spaceflight revealed inflammatory markers driving muscle and bone metabolism, which not only demonstrated the adverse impacts of microgravity during flight, but also the impacts of returning to gravity (Gertz et al., 2020).

Spaceflight-induced muscle loss, whether actual flight or ground analogs, bears similarities to catabolic patients undergoing metabolic breakdown. This is a critical and confounding issue because inadequate energy intake will lead to these same effects. Given that many, if not the vast majority, of astronauts don't meet energy intake requirements and lose body mass, the question remains to what degree does spaceflight in and of itself causes loss of muscle mass and performance, or whether these findings are confounded by inadequate dietary intake. This metabolic breakdown and energy intake are heavily related to mitochondrial biogenesis, and it has been reported that mitochondrial dysfunction plays a central role with this process (Wallace, 2005). It has even been reported that the interplay between oxidative damage and mitochondrial dysfunction can lead to muscular degeneration in the clinic (Civitarese et al., 2007).

Exercise is the most obvious countermeasure for muscle, bone, and cardiovascular health (Lang et al., 2017b). However, on Russian space station Mir flights, crewmembers differed significantly with respect to in-flight exercise frequency and intensity, but loss of leg muscle volume was almost 20% in all subjects (LeBlanc et al., 1996). A second-generation treadmill along with the Advanced Resistance Exercise Device (ARED) were launched to the ISS in 2008, which showed success at increasing lean body mass (Smith et al., 2012; Smith et al., 2014).

Bone Loss

Bone loss during spaceflight remains a leading concern for crew health because of multiple associated risks, including the risk of developing renal stones during the mission and an increased risk of bone fracture after flight (Smith et al., 2015b; Sibonga et al., 2020) (Figure 3A). The bone loss experienced during spaceflight is primarily from weight-bearing bones, and on average about 1%–1.5% of total bone density is lost per month of spaceflight (Sibonga et al., 2017). This is similar to the amount of bone lost over a year by a postmenopausal woman on Earth. Individual variability is a key characteristic of spaceflight-induced bone loss (Sibonga et al., 2017) and could provide insight to find a means to mitigate this loss. That is, astronauts can be evaluated to determine what they did differently that caused them to lose more (or less) bone than did other astronauts (e.g., exercise, diet).

A metabolically active tissue, bone is constantly undergoing turnover through breakdown (resorption) and formation processes. Bone resorption, assessed primarily through excretion of collagen crosslinks, is increased during spaceflight (Smith et al., 2005; Smith et al., 2015b), regardless of exercise countermeasures. Bone formation, on the other hand, is typically unchanged or decreased during spaceflight without effective exercise-induced bone loading (Smith et al., 2005). Together, increased levels of bone resorption and decreased or unchanged levels of bone formation during spaceflight yield an overall negative calcium balance and result in bone loss and increased risk of developing kidney stones (Smith et al., 2015a).

Resistance training utilizing the ARED aboard the ISS has allowed astronauts to maintain bone mineral density (Smith et al., 2012; Smith et al., 2014), although questions and concerns remain regarding bone architecture, turnover, qualitative strength impacts, and drug therapeutics (Leblanc et al., 2013). Many nutrients have been studied for their potential to mitigate bone loss, including calcium (Zorbas et al., 2008), omega-3 fatty acids, sodium, protein, potassium, and vitamin K (Smith et al., 2015b). Evidence exists that radiation, oxidative stress, and bone health are also related (Tian et al., 2017).

Immune Dysfunction

Spaceflight induces a complicated pattern of immune dysfunction in astronauts (Figure 3A). This dysfunction primarily consists of reduced T and natural killer (NK) cell function, changes in pattern-recognition system of monocytes and granulocytes, and changes in cytokine levels. These manifest in Th1/Th2 cytokine balance being shifted toward Th2-cell mediated immunity and a mild persistent inflammation, all resulting in the subclinical reactivation of latent herpesviruses (Crucian et al., 2014b; Mehta et al., 2017; Ponomarev et al., 2017). Adverse clinical events in select crewmembers include mild infectious disease, atypical allergy, and atopic dermatitis (Crucian et al., 2016). A similar type of immune dysfunction has been observed in animal models of spaceflight and ground analogs (Tahimic and Globus, 2017), astronaut blood (Barrila et al., 2016), and space-flown mouse splenocytes (Baqai et al., 2009). Observations from spaceflight and ground simulation studies indicate that mechanical forces can modulate the stimulus response thresholds of human T cells (Hauschild et al., 2014), which could partially explain the spaceflight effects.

In addition, spaceflight, post-flight reacclimation to Earth, and simulated deep-space radiation can induce mild chronic

systemic inflammation. Inflammation-associated transcriptomic signatures were reported by the Twins Study (Garrett-Bakelman et al., 2019), consistent with mild increases in inflammatory cytokine levels from other studies on astronaut plasma (Crucian et al., 2014b; Gertz et al., 2020). Similarly, simulated GCR also disrupts immune functions in animal models, increasing cardiovascular inflammation (Boerma et al., 2015) and neuroinflammation with associated neuronal damage (Acharya et al., 2019; Krukowski et al., 2018).

Notably, continuous immune studies spanning from the construction of the ISS to now indicate that more recent orbital crews could have improved immune status (Crucian et al., 2020), which has been ascribed to the deployment onboard the ISS of certain behavioral, exercise, and dietary countermeasures. An international team recently published a countermeasure protocol specific for maintaining immune competence during deep-space missions, consisting of nutritional supplements, specific exercise regime, stress relieving techniques, and pharmacological interventions (Makedonas et al., 2019). Further countermeasures could be developed based on the similarities between spaceflight-mediated immune dysfunction, immune deficiency, and autoimmune disorders (Crucian et al., 2014b; Mehta et al., 2017), which could enable drug repurposing from terrestrial diseases for spaceflight use.

Increased Liver Disease and Lipid Dysregulation

Evidence has recently begun accumulating highlighting key liver-related changes due to spaceflight. ISS, shuttle, and microgravity ground-analog animal livers presented with damage contributing to increased lipid and fatty-acid processing (Beheshti et al., 2019; Jonscher et al., 2016). These changes could lead to fibrosis and non-alcoholic fatty liver disease (NAFLD). da Silveira et al. (2020) demonstrated that increased levels of cholesterol (low- and high-density lipoprotein) were reported in astronauts during spaceflight (a total of 180 days), with levels still remaining elevated up to 30 days post-flight. This finding is consistent with increased plasma glucose levels and diabetogenic responses in astronauts during spaceflight (Tobin et al., 2002), which have been associated with increased risk of liver damage and NAFLD in the clinic (Xia et al., 2019) and fatty acid metabolism (Schmidt et al., 2020). In addition, spaceflight mitochondrial dysfunction (da Silveira et al., 2020) and mitochondrial damage due to increased ROS could induce lipid accumulation and also could lead to the development of NAFLD (Paradies et al., 2014). Together, these results indicate the importance of countermeasure development to mitigate potential liver damage for deep-space missions.

Circadian Rhythm Dysregulation

A distinct aspect of the confined and closed spacecraft environment is disrupted light-dark cycles. For example, the sunrise-to-sunrise interval on the ISS is 90 min. In addition, intervals of intense activity on spacecraft (“slam shifts” such as surrounding launch, docking, spacewalks) could expose crewmembers to the equivalent of shiftwork and deprive them of sleep. Circadian rhythms are present in processes ranging from molecular (e.g., gene expression) to systemic (e.g., cortisol levels) to behavioral (e.g., mood), all of which tend to be entrained by the light-dark cycle and might be disrupted by non-24-h periodicity or the presence of conflicting signals.

Circadian rhythm alterations in space have been found in a variety of model organisms. Alterations in circadian rhythms and gene expression in spaceflight have been described in *Drosophila melanogaster* (Ma et al., 2015). Space-associated body-temperature rhythm changes have been found in non-human primates in Cosmos missions (Fuller et al., 1996) and in rats during Neurolab missions (Fuller et al., 2003). Altered expression of circadian-associated genes and pathways have been identified in rodent spaceflight samples, including skeletal muscle (Allen et al., 2009) and liver tissues (Beheshti et al., 2019). Finally, alterations in circadian behavioral rhythms have been documented in astronaut gene-expression changes (Garrett-Bakelman et al., 2019), other astronauts (Flynn-Evans et al., 2016), and in shift-working flight controllers on Earth (Mizuno et al., 2016).

However, because a significant proportion of proteins or transcripts in a given tissue could exhibit diurnal or circadian changes (Zhang et al., 2014a), with single time point samples it is not possible to discriminate between differences in circadian phase, misalignment, reduction of amplitude, or other disruptions to the circadian system (Braun et al., 2018). Lacking data on the sleep-wake patterns at the time of sample collection, data interpretation is limited (Skene et al., 2018). Further studies with model organisms, including comparisons of samples collected at different times of day, or longitudinal samples (e.g., Twins Study), are needed to understand the nature of spaceflight’s effects on circadian rhythms.

Circadian rhythms are widely recognized to influence numerous health conditions, including many pertinent to spaceflight (Figure 3A). Circadian disruption has been found to disturb bone health (Song et al., 2018), metabolism (Panda, 2016), mitochondrial function (Sardon Puig et al., 2018), and inflammation and immune system function (Gachon et al., 2018). In addition, sleep and circadian organization are critical for gastrointestinal function (Godinho-Silva et al., 2019). Both experimentally induced circadian (Deaver et al., 2018) and sleep disturbances (Maki et al., 2020) can alter the normal gut microbiome, and the gut microbiome itself exhibits circadian or diurnal rhythms (Kuang et al., 2019). Given the circadian clock system’s function in temporal coordination of nuclear, mitochondrial, cellular, and systemic processes, circadian rhythms could be a common denominator connecting many of the features for spaceflight biology to health risks.

Space-Associated Neuro-ocular Syndrome

A shift of fluids from the lower to upper body occurs almost immediately upon exposure to spaceflight (Thornton and Rummel, 1977). The headward shift of about 2 L of fluid can have an effect on multiple systems, including increased cardiac output (Norsk et al., 2015) and gastrointestinal function (Lane et al., 1993). These contribute to congestion affecting perception by the special senses of taste and smell and could be a contributing factor to Space-Associated Neuro-ocular Syndome (SANS) (Figure 3A).

SANS is characterized by optic disc edema, choroidal and retinal folds, flattening of the posterior region of the sclera, and

hyperopic refractive error shifts (Lee et al., 2020). Although the precise mechanism is not yet fully understood, there are several published theories, some of which suggest an involvement of a cephalad fluid shift (Lee et al., 2020).

Elevated intracranial pressure (ICP) has been postulated to be a contributing factor to SANS, but so far there are no supporting data for this proposed mechanism, and astronauts typically report few or no symptoms (e.g., headaches) associated with elevated ICP (Laurie et al., 2020; Lawley et al., 2017). On the other hand, both spaceflight and strict head-down tilt bed rest have documented cases of optic disc edema (Laurie et al., 2020), and altered one-carbon metabolism has been associated with ocular changes in both of these models (Zwart et al., 2019). Recent studies indicate that some of the biology that might be responsible for SANS is related to the mitochondrial dysfunction and oxidative stress from systemic responses (da Silveira et al., 2020).

It is important to note that not all crewmembers who have flown long-duration missions develop SANS (Laurie et al., 2020), and it is more common in men, so the cause is likely multifactorial. For example, the impact of a cephalad fluid shift could be different between crewmembers due to genomic differences (Zwart et al., 2016; Garret-Bakelman et al., 2019). Genetic differences in one-carbon metabolism, combined with other factors such as fluid shifts, radiation exposure, CO_2, higher body-mass index, or a diet low in B vitamins could all contribute to endothelial dysfunction, which could contribute to leakier vessels and edema (Smith and Zwart, 2018). However, further studies are needed to unravel the extent to which these genomic and environmental factors contribute to SANS development.

Predicting Health Risks with Precision Analysis on Multi-omics Platforms

Although many of these systemic and physiological health risks of spaceflight have been well documented, much is left to be discovered. As long-duration, deep-space missions to Mars and farther are planned, it is uncertain how prolonged exposure to the space environment will alter the known physiological responses. Also, individual sensitivity to spaceflight stressors can greatly modulate biological responses, depending on genetic, demographic, and lifestyle variabilities. Thus, it is critical to establish and utilize a multi-omic approach as outlined in Figure 4 to further study these health risks and individual responses, including comprehensive monitoring of these molecular and cellular responses to enable personalized aerospace medicine, even on Mars (Nangle et al., 2020). In addition, multi-omics approachs have been used to predict risks of spaceflight hazards on individuals (Heuskin et al., 2016), as well as identify protection (Cortese et al., 2018). These models are made all the more accurate as data feed in from measurements and biomarker monitoring of the real astronaut environment through telemetry, dosimetry, microbiology, and even microgravity sequencing (McIntyre et al., 2016). To facilitate the success of these approaches in pushing the research forward with more orthogonal data sources, there are several publicly available resources and databases that include the NASA GeneLab platform and the NASA Life Sciences Data Archive (LSDA), as well as resources from other space agencies around the world (e.g., JAXA, ESA, ROSCOSMOS) (Box 1; Table 1).

NOVEL NUCLEIC ACID AND PROTEIN BIOMARKERS OF SPACEFLIGHT

Here we summarize recent discoveries regarding nucleic acids and proteins that are present in the blood as circulating factors and are involved in the regulation and measurement of spaceflight-relevant biological features (Figure 2). These circulating molecules include microRNAs, exosomes, and genetic markers for clonal hematopoiesis (Malkani et al., 2020; Paul et al., 2020; Wuu et al., 2020). They can be utilized as biomarkers to assess health risks during spaceflight and serve as potential targets for developing novel countermeasures against the systemic and physiological health risks (Figures 1 and 3A).

MicroRNAs as Key Regulating Factors of Responses to Spaceflight Stressors

MicroRNAs miRNAs are one type of small non-coding RNAs that are approximately 22 nucleotides (nt) in length and are involved in post-transcriptional gene silencing and modulation (Gebert and MacRae, 2019). miRNAs are found in every organism including mammals, invertebrates, microbes, viruses, and plants, and they are highly conserved between species (Moran et al., 2017). In addition, miRNAs are stable and functional in practically all bodily fluids (Silva et al., 2015). It has been reported that 10% of miRNAs found in bodily fluids are packaged in exosomes, and the remaining 90% of miRNAs are free floating in fluids with other proteins such as Ago2 (Silva et al., 2015). A recent review by Vanderburg and Beheshti (2020) on the importance of miRNAs with spaceflight described how miRNAs can impact every aspect of biology related to the space environment in many organisms, including mammals, microbes, and plants. Here we describe how circulating miRNAs can both respond to and affect the features of spaceflight biology based on recent publications not covered in that review.

In general, there have been limited studies with circulating miRNAs in spaceflight biology, but there is increasing evidence of their major regulatory role. Recently, Malkani et al. demonstrated that circulating miRNA signatures consisting of 13 miRNAs were associated with health risks induced by the space environment (Malkani et al., 2020). These miRNAs were linked to biological pathways involved with mitochondrial regulation and oxidative damage through multiple different factors, such as TGFβ1 and mTOR, known to induce mitochondrial oxidative phosphorylation and ROS production (Abe et al., 2013). In addition, these miRNAs were correlated with fatty-acid metabolism and biosynthesis, which is involved with mitochondrial dysfunction, oxidative damage, and DNA damage. A subset of the miRNAs was also shown to be involved with inhibition of angiogenesis and cardiovascular functions (Wuu et al., 2020), which are affected by several of the features of spaceflight biology (Figure 2).

Similarly, Paul et al., (2020) has recently demonstrated how circulating miRNAs from mice exposed to simulated space radiation (an acute dose of 0.5Gy of simulated GCR) and simulated

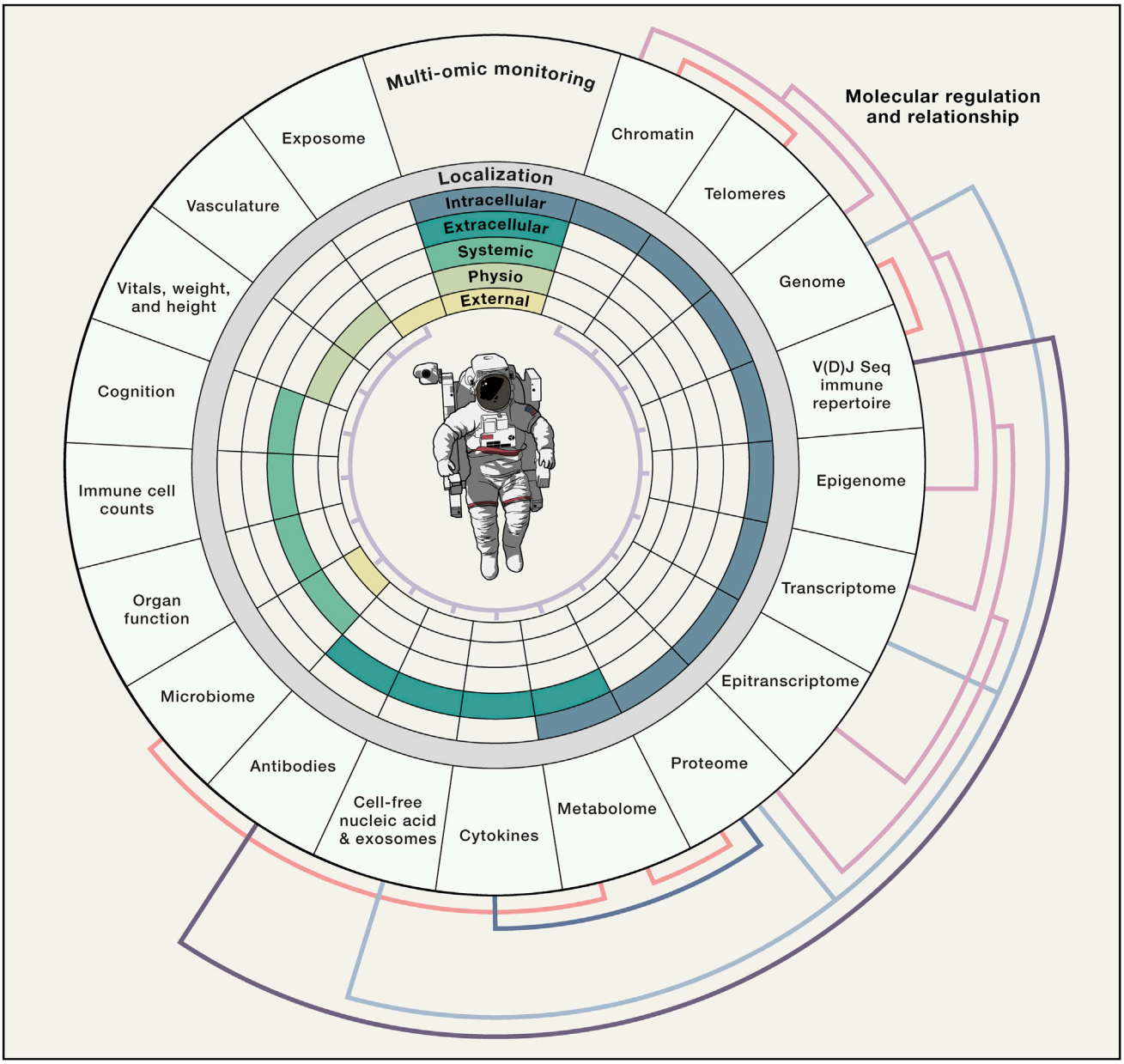


Figure 4. Multi-omics Monitoring Platform

Schematic of a multi-omics approach to monitor astronauts. Each -omic has its own level of localization (extracellular, intracellular, physiological, systemic, or external) and provides a unique insight that can be utilized for regular monitoring and follow-up. However, the molecular interactions highlighted in-between the different -omic data demonstrate the need to integrate all these measurements into one platform for the most comprehensive characterization and precise monitoring of an astronaut's health.

microgravity are involved with immune suppression, which again is associated with several features of spaceflight biology. MiRNAs have been shown to be both directly and indirectly involved in microbiome shifts, epigenetic changes, and DNA damage, with gut microbiome and miRNAs changes during spaceflight indicating that miRNAs have a role in determining microbial profiles (Vanderburg and Beheshti, 2020). Clinical cancer and radiotherapy studies have shown the association of miRNAs with chromosome aberrations and the relationship to space radiation. Interestingly, additional clinical data have shown how miRNAs can impact telomere function and structure inducing chromosomal instability (Dinami et al., 2014), which can warrant further investigation of how microgravity affects telomere integrity. Overall, circulating miRNAs can be a key factor to consider in systemic regulation for the biological responses to spaceflight, an optimal biomarker indicating onset of health risks, and potentially utilized for countermeasures.

Box 1. Spaceflight Biological Data and Biospecimen Resources

NASA data and non-human biospecimens are searchable and available through the NASA LSDA (Table 1). The LSDA provides information from decades of spaceflight research involving human, microbe, plant, and animal subjects. LSDA at NASA Johnson Space Center (JSC) is responsible for collecting and granting access to human research data. Similarly, the NASA Lifetime Surveillance of Astronaut Health (LSAH) at JSC collects and grants access to clinical and occupational health monitoring data from astronauts through LSDA. Data from plant investigations are archived through the LSDA at NASA Kennedy Space Center. The final node of LSDA, the Ames Life Sciences Data Archive (ALSDA) at NASA Ames Research Center (ARC), archives data and biospecimens from mammalian and microbial experiments and curates the biorepository within NASA's Institutional Scientific Collection at ARC, containing over 32,000 mostly mammalian biospecimens from predominantly ISS, shuttle, and ground-based investigations. The NASA Space Radiation Laboratory at Brookhaven National Laboratory exposes rodents to simulated space radiation with 150,000 rodent tissues available for request. NASA Planetary Protection engineers have monitored bioburden on spacecraft and associated surfaces since the Viking I mission in 1975. All isolates (~6,000) are archived in the Jet Propulsion Laboratory's microbial collection. The NASA GeneLab at ARC collects genomic, transcriptomic, proteomic, and metabolomic data from any species, including plants and microbes. It is a multi-omics repository and a platform for collaborative open-science bioinformatic approaches. The Astrobiology and Habitable Environments Database is under development at ARC as a repository and productivity platform for the storage, discovery, and analysis of data relevant to the field of astrobiology (the evolution and distribution of life in the universe). Internationally, the JAXA stores over 600 biospecimens remaining from spaceflight at the JAXA Tsukuba Space Center. JAXA is also establishing an omics-based repository in collaboration with the Tohoku Medical Megabank (ToMMo), called the JAXA-ToMMo Integrated Biobank for Space Life Science. This effort will link molecular mechanisms discovered in JAXA ISS experiments to large-scale human studies. In Europe, the Erasmus Experiment Archive is a database of European Space Agency-funded or co-funded experiments covering a wide range of scientific areas from the past 60 years. Roscosmos State Corporation for Space Activities, along with the Institute of Biomedical Problems (IBMP) of the Russian Academy of Sciences, has for decades developed medical and life-support systems, along with conducting spaceflight life science experiments. A long-standing successful series of biosatellite "BION" missions has resulted in many life science experiments being conducted, including international collaborations. Work is underway by IBMP for further data sharing and biospecimen sharing.

Exosomes and Cell-free DNA: Nanocarriers of Cellular Stress

Exosomes are nanovesicles secreted by most cell types as a means of intercellular communication through long-distance circulation in most biological fluids (Yáñez-Mó et al., 2015). The exosomal cargo is encapsulated in a phospholipid bilayer and comprises nucleic acids (including miRNAs), as well as proteins and lipids derived from the cell of origin and delivered to target cells (Pariset et al., 2017). The content of exosomes depends on the state of the donor cell and is modulated by environmental stressors, such as space radiation and microgravity. Spaceflight experiments showed hyper-production of exosomes (Bradamante et al., 2018) and enhanced exosome-mediated mRNA transfer (Li et al., 2018). Specifically, ionizing radiation triggers increased exosome release (Jella et al., 2014) and cellular uptake (Hazawa et al., 2014). Exosomes from irradiated cells are involved in radiation bystander effects through the synergic effect of miRNAs (Abramowicz et al., 2020), as well as proteins (Jelonek et al., 2015). Radiation-induced changes in the exosomal cargo induce chromosomal and genomic instabilities (Du et al., 2020), oxidative stress (Jella et al., 2014), inflammation (Szatmári et al., 2019), apoptosis (Mrowczynski et al., 2018), and migration enhancement in recipient cells (Mutschelknaus et al., 2017). Under simulated microgravity, osteoclast-derived exosomes were involved in the inhibition of osteogenic proliferation and differentiation (Huyan et al., 2019). Recent data from the Twins Study exosome profiles (Bezdan et al., 2020) showed brain-related peptides unique to the post-flight exosomes, as well as many other shifts after flight. Thus, the role of exosomes in the biological response to spaceflight should be carefully considered to better understand radiation and microgravity-induced health effects, especially with longitudinal data generated before, during, and after flight.

Somatic Mosaicism and Clonal Hematopoiesis

Beginning prenatally, inherent faults in the process of normal DNA replication result in accumulation of mutations in tissues and organs that appear otherwise healthy (Blokzijl et al., 2016; Vattathil and Scheet, 2016). Most of these mutations are neutral and thereby fail to confer a selective advantage under normal physiological conditions (Martincorena et al., 2017). However, as people age, clock-like mutational processes produce an increased repertoire of somatic mutations and thereby genetic diversity within each individual (Alexandrov et al., 2015). Also, deamination of 5-methylcytosine can lead to an altered "epigenetic clock." Thus, aging cells become a mosaic of mutant cells and epigenetic drift that can individually respond to selective pressure.

For instance, in the hematopoietic system, blood cells with advantageous mutations propagate faster than their non-mutated counterparts in a process called clonal hematopoiesis (CH). These mutations confer both an oncologic risk (Desai et al., 2018), as well as additional cardiovascular risks (Calvillo-Argüelles et al., 2019). Overall, CH has been shown to be associated with all-cause mortality (Jaiswal et al., 2014). The first CH characterization of former astronauts (Trinchant et al., 2020) indicates evidence of CH mutations. Of note, the CH observed in the astronauts studied so far appeared nearly two decades younger than the mean age in which these mutations are typically detected, suggesting certain factors of the space environment could accelerate this process, but more controls and context will be needed across larger cohorts to be sure. Indeed, longitudinal monitoring of CH and acquired genetic risk should be considered important metrics for further understanding the impact on biology due to spaceflight, along with evaluating the overall health and disease risk of astronauts.

Table 1. Data and Biospecimen Resources

Data Repositories	Type	Research or Payloads	Factors	Species	Volume	Resource Links
NASA Lifetime Surveillance of Astronaut Health (LSAH)	human clinical	astronaut corps and payload specialists	occupational hazards, lifetime risks	human	50 categories of data collected	https://lsda.jsc.nasa.gov/Request/dataRequestFAQ; https://lsda.jsc.nasa.gov
NASA Life Sciences Data Archive (LSDA)	mission or experiment or biospecimen metadata, experiment assay data, hardware specifications, environmental telemetry, mission science reports, imagery	Apollo, BION, Biosatellite, Cosmos, Gemini, Mercury, MIR, Skylab, Shuttle, ISS, SpaceX, bed rest, Human Research Facility, NEEMO, ground analogs	various	human, monkey, rat, mouse, fish, frog, fly, nematode, quail, plant, microbe, others	Dataset Entries 1,239 human, 748 rodent, 54 plant, 48 microbial Experiment Entries 1,490 human, 50 monkey, 613 rodent, 143 plant, 74 microbial	https://lsda.jsc.nasa.gov/Request/dataRequestFAQ; https://lsda.jsc.nasa.gov
NASA GeneLab	-omics	shuttle, ISS, parabolic flight, NASA Space Radiation Lab (NSRL)	microgravity, radiation, altitude, hypergravity	animal, microbial, plant, human (non-astronaut)	278 studies (322 datasets)	https://genelab.nasa.gov
NASA Astrobiology Habitable Environments Database (AHED; under development)	chemistry, biology, astrophysics, geology, planetary science	field, lab, computational modeling, missions	various	various	N/A	https://www.nasa.gov/ames/ahed
JAXA-ToMMo Integrated Biobank for Space Life Science (ibSLS)	-omics	ISS	microgravity, artificial gravity	mouse	5 ISS missions	https://ibsls.megabank.tohoku.ac.jp/
European Space Agency (ESA) - Erasmus Experiment Archive (EEA)	research descriptions	ISS, space shuttle, Mir, Salyut, Skylab, Apollo-Soyuz, Apollo, Foton, Bion, Eureca, sounding rockets, parabolic flights, drop tower, Concordia, bed rest, simulation studies	highly various	human, microbes, plants, cells, others	~2,550 experiment entries on life sciences. no datasets	http://eea.spaceflight.esa.int
Roscosmos State Corporation for Space Activities and Institute for Biomedical Problems (IBMP; under development)	raw data, experiment annual/phase/final reports	ISS, Biosattelites (BION), Foton, ground analogs	low-earth orbit spaceflight, deep spaceflight, artificial gravity	human, animal, fly, microbe, fungi, cells	N/A	under development

(*Continued on next page*)

Table 1. *Continued*

Biospecimen Repositories	Type	Research or Payloads	Factors	Species	Volume	Resource Links
NASA Institutional Scientific Collection (ISC) at Ames Research Center, and Biospecimen Sharing Program (BSP)	animal, microbial	Cosmos, shuttle, ISS, rodent research, ground analog studies	microgravity, hindlimb unloading, centrifuge, partial weight-bearing, others	mouse, rat, quail, microbes	32,000 specimens	https://lsda.jsc.nasa.gov/Request/dataRequestFAQ; https://www.nasa.gov/ames/research/space-biosciences/isc-bsp
ISS Microbial Analysis	monitoring within ISS	ISS	air, surface, water	microbes	27 ISS missions	https://lsda.jsc.nasa.gov/Request/dataRequestFAQ; https://lsda.jsc.nasa.gov/Experiment/exper/13823#data
HRP Space Radiation Element's NASA Space Radiation Laboratory (NSRL) Biospecimens	mouse, rat	ground research	energy, dose, particle type	mouse, rat	>150,000 specimens	https://lsda.jsc.nasa.gov/Request/dataRequestFAQ; https://www.nasa.gov/hrp/elements/radiation/about
NASA Planetary Protection	spacecraft bioburden	Viking, Pathfinder, Odyssey, exploration rovers, Phoenix, Mars Science Laboratory, InSight, Mars 2020	Jet Propulsion Lab (JPL) and Kennedy Space Center (KSC) Clean Rooms	microbes	5,674 cultures	https://planetaryprotection.jpl.nasa.gov/biological-materials-archive
JAXA Biorepository at Tsukuba Space Center	animal, microbial	Mouse Habitat Unit, Aquatic Animal Experiment Facility	microgravity, on-orbit centrifugation	mouse	600 specimens	https://iss.jaxa.jp/kibouser/provide/mhu/71891.html
Institute for Biomedical Problems (IBMP; under development)	human, animal, microbes, cells, fungi	ground research	simulated microgravity isolation, radiation, hypomegnetic conditions	human, animal, microbes, cells, fungi	N/A	Under development

Summary table of available data and biospecimen repositories for the further characterization and study of the hallmarks of space biology.

FUTURE DIRECTIONS IN SPACEFLIGHT BIOLOGY RESEARCH

In summary, the battery of known, spaceflight-mediated health hazards collectively affect multiple biological systems, spanning space radiation, microgravity, confinement/isolation, a hostile/closed environment, and distance from Earth. These hazards provoke recurrent biological features of spaceflight, including oxidative stress, DNA damage, mitochondrial dysregulation, epigenetic/regulatory changes, telomere-length dynamics, and shifts in host-microbe interactions. Health impairments span multiple physiological systems from cardiovascular, central nervous, musculoskeletal, hepatic, and immune, as well as increased cancer/mutational risk and disrupted circadian rhythms. Although significant advances have been made in the past decade to understand these hazards and health risks, additional research is required to enable safer human space exploration beyond LEO, including lunar, Mars, and deep-space missions. Also, applied research is required to assist with countermeasure development and efficient biomarker monitoring for astronaut health. The fields of space biology and aerospace medicine will benefit from multi-omic, longitudinal profiling that capture the combined effects from exposures across multiple hazards and interactions between multiple biological features and organ systems. Each feature of spaceflight biology alone can only provide limited insights, but a collective analysis across multiple models and human studies can lead to a more comprehensive understanding of physiological and human health-related impacts of the space environment. This concept underscores the importance of utilizing a multitude of space and ground analog models and integrative analyses outlined in Figure 1 as well as multi-omics approaches (Figure 4) to study these features (Figure 2).

Sending experiments and organisms into space is a costly endeavor (~$10,000 USD/kg). To maximize scientific return, raw data and metadata from space-flown and ground analog experiments, as well as biospecimens not required by Principal Investigators, are deposited in a variety of platforms. Data and biospecimen repositories are publicly searchable with much data being open access, although certain portions are available only by request (Table 1). Altogether, open access to these resources enable researchers to seek new discoveries through fundamental research, applied research, bioinformatics, and translational science. Data reuse from spaceflight investigations and monitoring holds promise in enabling spaceflight health risk modeling, semi-autonomous medical-support systems, establishing data-science-assisted knowledge bases for health-related diagnostic support, and analyzing adverse health outcomes across concurrent spaceflight hazards. Further descriptions of the data and biospecimen resources available are found in Box 1.

In 2004, NASA released a "Vision for Space Exploration" outlining the goal of human exploration of Mars (NASA, 2004). It suggests that all efforts be designed with Mars as the intended target, including life-science discoveries and mission analogs. Current government-funded research, typically involving international collaboration, focuses on human exploration of the Moon as a staging ground in conjunction with research and preparations for the underlying goal of Mars. Figure 3B depicts some of the planned missions encompassing NASA, international, and commercial ventures related to health risks, which also can build on the continuous planned missions that occur on the ISS throughout the years ahead. Although a number of these planned missions are of a longer duration and are farther away from Earth, there remains a contribution from LEO toward supporting lunar and Mars missions. Of note, biological life-science missions on, or orbiting, the Moon or Mars (e.g., Mars Base Camp) will play a key role in the scientific pursuit of fully elucidating GCR and space radiation as a deep-space health risk. Lunar-oriented experiments and laboratory platforms will be distant from the radioprotective effects conferred by the Earth's magnetosphere. There are already extensive plans that have been laid out for long-term settlement on the Moon and Mars (Nangle et al., 2020), and most missions revolve around these types of scientific efforts and human exploration.

Finally, the advent of commercialization and tourism in LEO creates opportunities for research that is entirely unprecedented in spaceflight biology research, namely larger sample sizes, more datasets, and study populations of a greater diversity. Additionally, Earth-based human spaceflight analogs are promising areas of future research, because they are cheaper and provide results more quickly than spaceflight missions. Challenges facing Earth-based analog studies include feasibility and fidelity, as reported in Keeton et al., 2011. These challenges extend to model organism analogs too, and they are important when pursuing biological features of spaceflight discovery. To conclude, the features of spaceflight biology we outlined are foundational but are likely only the beginning of what will be discovered. Understanding the health implications from the features and developing effective countermeasures and health systems are key steps in enabling humanity to reach the next stage of space exploration.

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Correction

Fundamental Biological Features of Spaceflight: Advancing the Field to Enable Deep-Space Exploration

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We have identified that in our original article, we inadvertently directed readers to https://jmorp.megabank.tohoku.ac.jp/202008/downloads/legacy/ in Table 1, regarding the JAXA-ToMMo Integrated Biobank for Space Life Science data repository. Instead, this URL should direct readers to https://ibsls.megabank.tohoku.ac.jp/. The URL has now been corrected online, and the authors apologize for any confusion it has caused.

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